UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which operator sends data to cout?
   A. <<
   B. =
   C. ==
   D. >>

2. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. p
   B. n
   C. t
   D. i

3. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Coding
   C. Testing
   D. Maintenance

4. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 246
   B. 1357
   C. 1234
   D. 2468

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, ii, iii, iv
   B. i, ii, iii, iv, v
   C. i, ii, iii, v
   D. i, iii, iv

6. Which structure chooses between two alternative paths?
   A. array
   B. struct
   C. if...else
   D. for

7. Which data type stores a whole number?
   A. float

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B. string
C. char
D. int

8. Which loop is guaranteed to execute its body at least once?
A. while
B. if
C. do...while
D. for

9. Which of the following declarations are valid?
i. int a[5];
ii. int b;
iii. int c[3] = {1, 2, 3};
iv. int d[];
A. i and ii
B. ii and iv
C. iii and iv
D. i and iii

10. Which values can a bool variable hold?
A. any integer
B. true or false
C. any character
D. any decimal

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(3 marks)

b) State THREE differences between a compiler and an interpreter.

(2 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(8 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(5 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(7 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(8 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

END OF QUESTION PAPER

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